

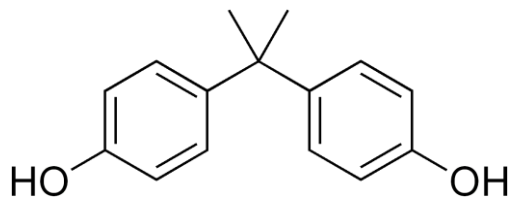
Bisphenols (雙酚類)

What are Bisphenols?

Bisphenol A (BPA) is a **precursor chemical** used in the production of plastics and resins.

It is commonly used to harden plastics.

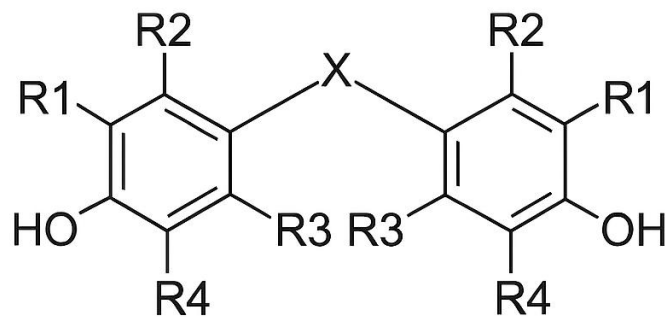
BPA 是一種**前驅單體**，常用來製造塑膠，環氧樹脂，塗層等，**常用於硬化塑膠**。



Bisphenol A (BPA)

BPA derivatives include Bisphenol S (BPS), Bisphenol B (BPB), Bisphenol F (BPF), etc. fulfil the WHO/IPCS (International Program Chemical Safety) criteria for **endocrine-disrupting substances** in the environment.

BPA 衍生物包括雙酚S (BPS), 雙酚B (BPB), 雙酚F (BPF) 等, 符合 WHO/IPCS 對環境中**內分泌干擾物質**的標準。



BPA derivatives such as BPS, BPB, BPF, etc are sharing the similar structure



Use of Bisphenols in the Textile Industry

BPA is used in the **production of polymers and resins** for plastic materials.

BPA用於製造塑膠材料的高分子和樹脂。

BPA serves as a **monomer** in various polymer manufacturing processes, including polysulfones, polyarylates, polyetherimides, unsaturated polyesters, benzoxazines, phenolic resins, and dental resins.

BPA作為單體存在於多種聚合物製造過程中，包括聚磺胺，聚芳烯酸酯，聚醚酰亞胺，不飽和聚酯，苯並茚，酚醛樹脂及牙科樹脂。

BPS and **BPF** in **textile auxiliaries** act as **color fixers for polyamide textiles, enhancing color longevity.**

紡織輔助劑中的 BPS 和 BPF 作為聚醯胺織物的固色劑，提升色彩持久度。

Synthetic tanning agents (aromatic syntans) enhance **leather stability** and protect **against microbial degradation.**

合成鞣製劑（芳香鞣）能提升皮革穩定性並防止微生物分解。

Syntans serve as leather tanning agents and **textile color fixers.**

合成劑可作為皮革鞣製劑及紡織品的固色劑。

Textile auxiliaries, particularly **aromatic syntans (based on naphthalene, melamine, and elastane), act as post-treatment agents for dye fixation.**

紡織輔助劑，特別是芳香合成酞（基於萘，三聚氰胺和彈性烷）作為染料固定的後處理劑。

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Bisphenol A (BPA) in US – Prop 65 Notices on BPA

BPA is a well-studied chemical, known to cause developmental and **reproductive harm** and linked to **breast cancer, prostate cancer**, metabolic disorders, diabetes, and numerous other serious health concerns. **It was added to the California Proposition 65 (Prop 65) list of chemicals in 2015.**

BPA 被列入 Prop 65 清單是在 2015 年, 因為它與發育性生殖毒性傷害, 也與多種疾病 (如乳癌 & 攝護腺癌) 有潛在的關聯

May 2022, the first known settlement for **Socks** made primarily of **Polyester with Spandex** which contained **Bisphenol A (BPA)** was finalized.

2022 年加州第一次針對一個襪類產品 (**Polyester + Spandex** 的襪子) 開罰, 因為它含有 **BPA**.

- **Reformulation: 1 part per million (ppm)** BPA as measured by the Test Protocol, **otherwise warning**.
- Cannot replace the BPA with any other phenols (such as Bisphenol S or BPS).

June 2023 and January 2024, consent judgment under **Prop 65** has been finalized for **athletic shirts and leggings** composed mainly of **Polyester with Spandex** that contain **Bisphenol A (BPA)**.

2023 年與 2024 年又出現兩個案例, 這次是針對 **Polyester + Spandex** 的運動服與緊身褲.

- **Reformulation limit** of no more than **10 parts per billion (ppb) = 0.01 ppm BPA**, **warning option is allowed**.
- Prohibits the substitution of BPA with other bisphenols (such as BPS).

BPA-based chemical compounds can also be found in **Polycarbonate bottles, food and beverage cans, thermal paper**, storage containers, **plastic sunglasses**, adhesives and coatings, as well as the **above-mentioned textiles**.

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Bisphenol A (BPA) in EU REACH

Bisphenol A is classified in the EU as a substance that:
Causes toxic effects on our ability to reproduce (Repr. 1B)
生殖毒性, 影響生育能力

May cause respiratory irritation (STOT SE 3) 呼吸道刺激
Causes serious eye damage (eye dam. 1) 嚴重眼睛損傷
May cause skin allergies (skin sens. 1) 皮膚致敏

BPA has been restricted as a substance on its own and in mixtures intended for consumer use in the EU since March 2018.
Its use in **thermal paper** has been restricted since January 2020.
BPA, Bisphenol B (BPB), 2,2-bis(4-hydroxyphenyl)-4-methylpentane and Bisphenol S (BPS) have been identified as **SVHCs**.

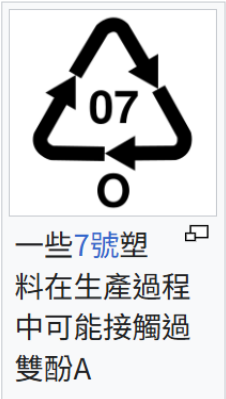


Recycle Plastic related

In general, plastics with plastic classification marks **1, 2, 4, 5, 6** are less likely to come into contact with BPA in production, while some, but not all, plastics with plastic classification marks **3 or 7** may be exposed to BPA in production.

Recycling logos and plastic classification logos can often be seen on plastic material products, with one of seven Arabic numerals from 1 to 7 printed in the center **The "7" stands for all "other plastics," including polycarbonate (sometimes marked with the word "PC" near the recycling logo) and epoxy, which are made from bisphenol A.**

PVC 3 and some plastics 7 may also contain BPA, as BPA is added to plasticizers as an antioxidant in the production of flexible PVC.



一般來講，塑膠分類標誌為 **1, 2, 4, 5, 6** 的 塑膠 不太可能在生產中與雙酚A接觸，而有一些（不是全部）塑膠分類標誌 為 **3 或 7** 的塑料生產中可能接觸到了雙酚A

塑膠材料製品上往往可以看到 循環再造標誌 和塑膠分類標誌，標誌中央印有包括 1 至 7 七個阿拉伯數字中的一個，其中的「**7**」代表所有的「其它塑料」，包括 聚碳酸酯（有時會在循環再造標誌附近註明“PC”字樣）和 環氧樹脂，它們的生產原料都是雙酚A

3 號（PVC） 和一些 **7 號** 塑料同樣可能含有雙酚A，因為在生產軟質 PVC 的過程中，雙酚A會作為 抗氧化劑 被添加到 塑化劑 中



Recycle Plastic related

Recycled Polyester, also known as RPET is created by melting down existing plastic and re-spinning it into new polyester fibers.

Mixed Recycling Streams - Sorting is never 100% perfect, and PC items may get mixed into PET recycling.

Heterogeneous Flake Composition - RPET flakes often come from multiple sources, making it difficult to guarantee uniform purity.

Cleaning & Processing Limitations - Even with rigorous washing and filtration, some BPA could remain.

Testing Bottlenecks - Since the contamination is at the flake level, testing individual batches of yarn is the only sure way to detect BPA.

And if the yarn contains BPA, then you have to either accept the impurities or trash the whole batch which is terrible from the environmental perspective.

回收聚酯，也稱為 RPET，是透過熔化現有塑膠並重新紡製成新的聚酯纖維製成。

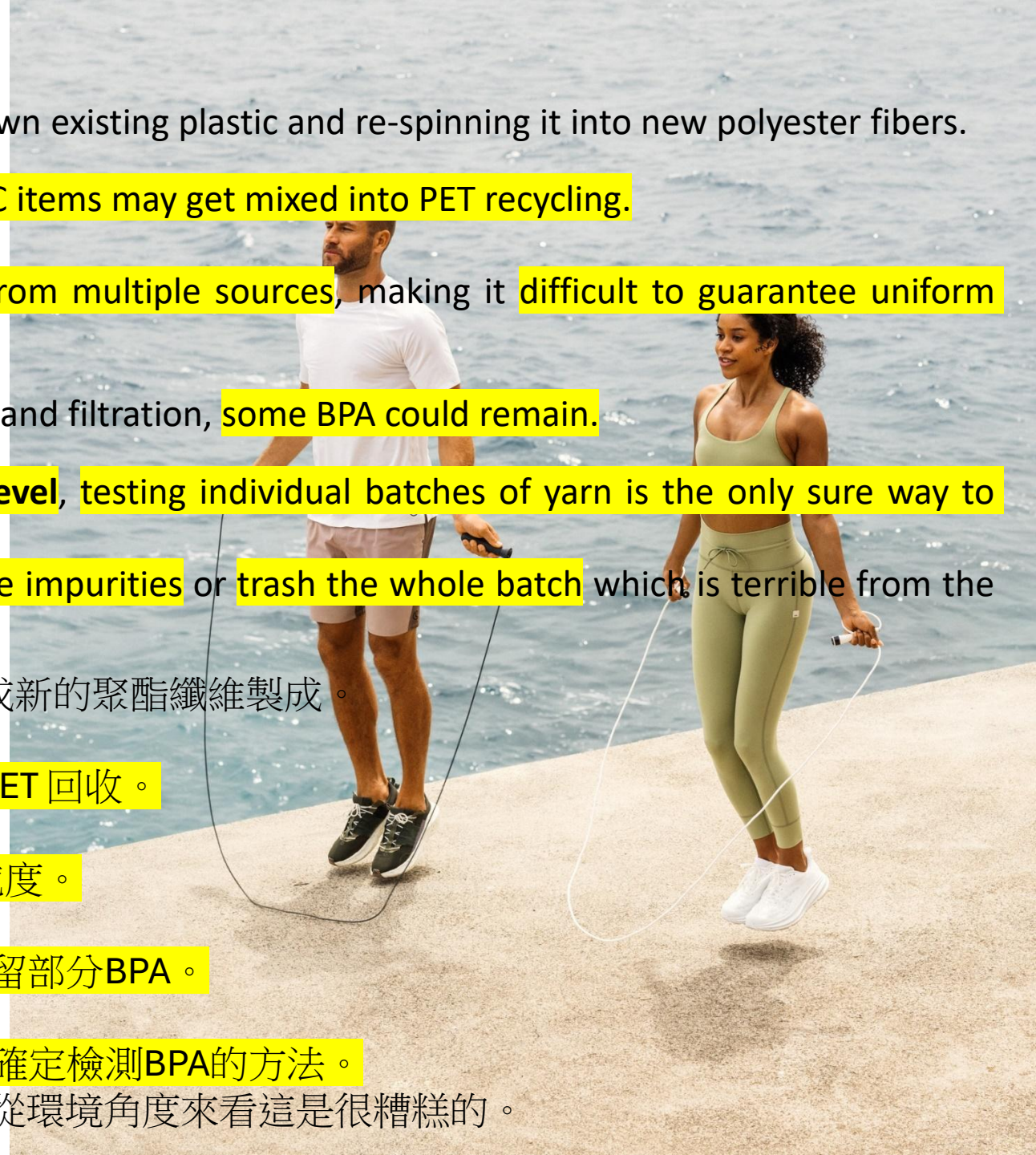
混合回收流 - 分類永遠不會百分之百完美，PC 可能會混入 PET 回收。

異質片狀組成 - RPET 片通常來自多種來源，難以保證均勻純度。

清潔與加工限制 - 即使經過嚴格的清洗和過濾，仍有可能殘留部分 BPA。

檢測瓶頸 - 由於污染發生在片狀層面，檢測單批紗線是唯一確定檢測 BPA 的方法。

如果紗線含有 BPA，那你就得接受雜質，或者整批都丟棄，從環境角度來看這是很糟糕的。



Compliance

Precautionary Awareness

- Build in at development stage

Product Evaluation

- Identify the Risk
- [Tracking Record](#)

Assessment

- Testing/CAP

Compliance

- Assurance

High Risk Materials

Bisphenol	Nylon/Spandex Recycled/non-recycled)	Polyester/Spandex (Recycled/non-recycled)	Plastic	Thermal Paper
BPA	●	●	●	●
BPS	●	●		●
BPF	●			

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Key Takeaway about Bisphenols

What are Bisphenols?

CAS Number	Substance
80-05-7	Bisphenol-A (BPA)
80-09-1	Bisphenol-S (BPS)
620-92-8	Bisphenol-F (BPF)
1478-61-1	Bisphenol-AF (BPAF)

May Be Found In:

- Polycarbonate bottles
- Food and beverage cans
- Thermal paper
- Storage containers
- Plastic sunglasses
- Impact resistant safety equipment
- Adhesives, coatings, trims
- Dye-fixing agents for polyamide textiles

Why Bisphenols are Restricted

- Numerous countries in the European Union, Americas and Asia have adopted restrictions on the use of BPA in infant products, namely baby bottles.
- Human exposure to BPA is pervasive. A national health survey conducted in 2003-2004, reported that 93% of Americans aged 6 years or older had detectable levels of BPA.
- BPA is an endocrine disrupter, associated with risks that may include metabolic changes, cardiovascular diseases, impact to reproductive systems, and others.
- At the manufacturing level, human exposure can result from inhalation or contact to the skin.
- At the consumer level, BPA exposure is a result of migration that occurs when BPA penetrates food or beverage from the lining of the container or containers made from BPA-containing plastics. Additional exposure routes can occur through leaching from dental fillings that contain BPA or contact with thermal paper.

Bisphenol-S has endocrine disrupting properties and is regulated in Food Contact Materials.

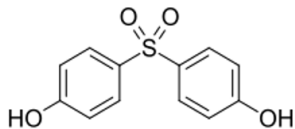
Bisphenol-F is suspected to have endocrine-disrupting properties similar to BPA.

Bisphenols are precursor chemicals used along with other chemicals to create some plastics and resins. They are commonly used to harden plastics.

Uses in the Supply Chain

Bisphenol-A (BPA) occurs in its pure form as white flakes with a faint phenol-like smell and is often used in the production of polycarbonate plastics and epoxy resins. Polycarbonate plastic is used across various product types for its durability, clarity, and shatter resistance, making it an important component in medical appliances or optical lenses.

Epoxy resins are used to prevent rust and corrosion (e.g. in the linings of food and drink cans). BPA is also known to be used in the development of dyes that are used with thermal paper (commonly seen in cash register receipts). BPA is also used in the production of flame retardants and in PVC production and processing. The other Bisphenols listed here have similar properties to BPA and can be used as alternatives to BPA in the supply chain. BPS may be used as an ingredient in dye-fixing agents for polyamide textiles.



AFIRM Restricted Substances List

CAS No.	Substance	Limits Component Materials in Finished Product	Potential Uses & Additional Information
	Bisphenols EN CN		
80-05-7	Bisphenol-A (BPA)	Textiles & Leather: 10 ppm Items intended to come in contact with the mouth: 1 ppm Other Materials: 1000 ppm	BPA may be used in the production of epoxy resins, polycarbonate plastics, flame retardants, and PVC. BPS may be used as a substitute for BPA for some specific uses, including in thermal receipt paper. BPS and BPF can be found in polyamide dye-fixing agents and in sulfone- and phenol-based leather synthetic tanning agents.
80-09-1	Bisphenol S (BPS)	Textiles: 200 ppm each Leather: 800 ppm each	BPA and BPS can be found in recycled polymeric and paper materials due to polycarbonate plastic and thermal receipt paper made with bisphenols entering waste streams.
77-40-7	Bisphenol B (BPB)	Limits will likely be reduced further in future revisions of the AFIRM RSL based on the best available technology and feasibility within the supply chain.	BPA, BPS, and BPB are included on the REACH SVHC list. Additional restrictions on the entire class of bisphenols are expected, with a revised restriction proposal forthcoming in the European Union.
620-92-8	Bisphenol F (BPF)	Other materials: 1000 ppm each	Important: Compliance with bisphenol (and other) limits in the AFIRM RSL does not prevent public or private enforcers from asserting that products violate California Proposition 65 warning obligations. AFIRM recommends testing relevant materials for bisphenols according to the Testing Matrix and working with suppliers to minimize residual concentrations or replace them with better alternatives where possible.

CURRENT 2025 AFIRM RSL → 2026

BPA:
Textiles & Leather: 10 ppm

Items intended to come in contact with the mouth: 1 ppm

Other Materials: 1000 ppm → 200 ppm

BPS, BPB, BPF:

Textiles: 200 ppm each

Leather: 800 ppm each → 500 ppm

Other Materials: 1000 ppm → 200 ppm



Bisphenols 					Reporting Limits
80-05-7	Bisphenol-A (BPA)	Items intended to come in contact with the mouth: 1 ppm Textiles & leather: 10 ppm Polycarbonate Materials: 100 ppm Other Materials: 200 ppm	BPA may be used in the production of epoxy resins, polycarbonate plastics, flame retardants, and PVC. BPS may be used as a substitute for BPA for some specific uses, including in thermal receipt paper. BPS and BPF can be found in polyamide dye-fixing agents used to dye nylon and in sulfone- and phenol-based leather synthetic tanning agents. Metal-complex dyes used on nylon may contain BPS.	Textiles: For precipitation, draw the extract to another container and add methanol or acetonitrile. This keeps the extraction process consistent.	
80-09-1	Bisphenol S (BPS)	Textiles & other materials: 200 ppm each Leather: 500 ppm each Limits will likely be reduced further in future revisions of the AFIRM RSL based on the best available technology and feasibility within the supply chain.	BPA and BPS can be found in recycled polymeric and paper materials due to polycarbonate plastic and thermal receipt paper made with bisphenols entering waste streams.	Extraction: 1 g sample/20 mL THF, sonication for 60 minutes at 60° C, then add methanol or acetonitrile for precipitation prior to analysis with LC/MS.	Leather: 10 ppm each
77-40-7	Bisphenol B (BPB)		BPA, BPS, and BPB are included on the REACH SVHC list. Additional restrictions on the entire class of bisphenols are expected, with a revised restriction proposal forthcoming in the European Union.	Leather: EN ISO 11936:2023	All other materials: 0.1 ppm for individual samples
620-92-8	Bisphenol F (BPF)		Important: Compliance with bisphenol (and other) limits in the AFIRM RSL does not prevent public or private enforcers from asserting that products violate California Proposition 65 warning obligations. AFIRM recommends testing relevant materials for bisphenols according to the Testing Matrix and working with suppliers to minimize residual concentrations or replacing them with better alternatives where possible. Washing nylon fibers can remove free BPS but sufficient wastewater treatment should be in place.	Other Materials: Extraction: 1 g sample/20 mL THF, sonication for 60 minutes at 60° C, then add methanol or acetonitrile for precipitation prior to analysis with LC/MS.	1 ppm for composite samples

Regulatory Compliance as Driving Force for eliminating Bisphenols from our Products

Region	Key Restrictions	Year Implemented	Additional Notes
European Union	Restriction proposal for bisphenols; limit 10 ppm; BPA banned in infant feeding bottles and packaging	2011, 2018, 2020	Candidate list includes BP A,B,S,F,AF; REACH BPA restriction
United States	FDA restricts BPA in baby bottles, sippy cups, infant formula packaging; California bans bisphenols in children’s products	2015	BPA listed in California Proposition 65; TSCA Inventory by EPA
Canada	BPA identified as toxic; ban on baby bottles containing BPA; prohibited in cosmetics	1999	Listed on Canadian Cosmetic Ingredient Hotlist
China	Prohibition of manufacturing, import, and sale of infant feeding bottles containing BPA	2011	Announced by Ministry of Health
United States	Prop 65 listing of BPA as reproductive toxin	2015	Added to California Proposition 65 list
United States	Settlement for socks with Polyester + Spandex containing BPA; reformulation limit 1 ppm	2022	Cannot replace BPA with other phenols (e.g., BPS)
United States	Prop 65 judgment for athletic shirts and leggings; reformulation limit 10 ppb = 0.01 ppm	2023, 2024	Prohibits substitution of BPA with other bisphenols
United States	BPA-based compounds found in polycarbonate bottles, food/beverage cans, thermal paper, storage containers, plastic sunglasses	Ongoing	Includes adhesives and coatings; above-mentioned textiles

Item	Type	Limit / Threshold	Units	Exposure Route / Product	Effective Year(s)	Notes	Source
BPA Safe Harbor (MADL)	Exposure limit (safe harbor)	3	µg/day	Dermal from solid materials	Effective Oct 1, 2016 (listed 2015)	Regulatory safe harbor limit; warnings required if exposure > 3 µg/day	OEHHA MADL for BPA (dermal)
BPA in Socks	Settlement reformulation	1	ppm	Socks (polyester + spandex)	May 2022	If >1 ppm, warning allowed; cannot substitute with other bisphenols (e.g., BPS)	SGS SG 083/22; CGC-22-598022
BPA in Athletic Shirts/Leggings	Consent judgment reformulation	10	ppb (0.01 ppm)	Athletic shirts/leggings (polyester + spandex)	Dec 2023 & Jan 2024	If >10 ppb = 0.01 ppm, warning option allowed; substitution with other bisphenols prohibited	CA AG settlement 2023-01853S2926; SGS SG 167/23

3 µg/day ≈ ~ 300ppm (estimated)
Depends on fabric type, contact area, exposure duration, and migration rate.
For Proposition 65 compliance, brands typically try to ensure BPA levels are well below 300ppm in wearable products.

Item	BPA	BPS
Prop 65 Status	Listed in 2015 (Repro Toxin)	Listed in 2023 (Repro Toxin)
MADL (Maximum Allowable Dose Level)	3 µg/day (dermal)	Not established
Estimated Safe ppm	~300ppm (under standard wear assumptions)	No definitive estimate, but conservative brands treat ~300ppm as precautionary limit
Warning Label	Required if exposure > MADL	Required if exposure is any, unless proven negligible

Case ID	Company	Product Type	Bisphenol Type	Notice Date	Settlement Date	Settlement Amount	Source Document Link
2021-02541	Multiple brands (e.g., Dollar General, Adidas)	Polyester-spandex socks	BPA	2021-10-13	2022-05-15	\$750,000	https://oag.ca.gov/system/files/prop65/settlements/2021-02541.pdf
2021-02177S1293	Fit For Life LLC.	Polyester-spandex socks	BPA	2021-09-01	2023-01-15	\$20,000	https://oag.ca.gov/system/files/prop65/settlements/2021-02177S1293.pdf
2023-01853S2926	Alo, LLC. & Athleta LLC.	Athletic shirts and leggings	BPA	2023-06-28	2023-09-29	Non-disclosed - Estimated US\$45,000	https://oag.ca.gov/system/files/prop65/settlements/2023-01853S2926.pdf
2024-00123	Levi's	Textile/Apparel	BPA	2024-01-23	2024-06-15	\$500,000	-

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Chemical	Classification	Safe Harbor Limit (MADL)*	Exposure Route	Notes
BPA	Reproductive Toxicant (since 2015)	3 µg/day (micrograms/day)	Dermal (skin contact from solid materials)	Applies to clothing/textile articles
BPS	Suspected Reproductive Toxicant (2023)	Not yet established	Pending assessment	Listed but no MADL yet published

*MADL = Maximum Allowable Dose Level

Reference: OEHHA Proposition 65 MADL list

<https://oehha.ca.gov/proposition-65/general-info/proposition-65-no-significant-risk-levels-nsrls-and-maximum-allowable-dose-levels-madls>

Notes: The exposure levels were assessed based on dermal contact through clothing, particularly relevant during activities involving sweating and friction.

No specific PPM (parts per million) data were publicly disclosed in testing reports.

All brands listed above failed to provide Proposition 65 warnings, which is required if exposure exceeds the safe harbor limit.

要將 3 微克/天 (µg/day) 的 BPA 曝露限值轉換為 ppm (parts per million), 需要一些假設, 因為 ppm 是針對「材料濃度」的單位, 而 µg/day 是「人體每日吸收量」的單位. 兩者不同, 換算必須基於: 接觸面積 (例如) 衣服接觸皮膚的面積, 接觸時間 (每天幾小時), BPA 從材料遷移到皮膚的比例 (皮膚吸收率), 材料中 BPA 含量 (ppm) 對應每日釋放量 (µg/day)

一般參考假設模型 (根據 CEH 常用評估方法) 假設每天穿著該衣物 10 小時, 衣服面積與皮膚接觸達 1,000 cm² 每 cm² 衣料釋放量為 0.01 µg/day 即: 總釋放量 = 1,000 cm² × 0.01 µg = 10 µg/day 若材料中的 BPA 含量為 1,000ppm, 可釋放 10 µg/day, 則估算: 3 µg/day ≈ 約 300ppm (粗略值, 依使用條件而變) 這表示: 若衣服中 BPA 含量超過 300ppm, 則可能會導致使用者曝露超過 Prop. 65 的 3 µg/day 限值.